RESCUE MIND

Team RescueMind is from **South Korea** and participates in the **Rescue Simulation**

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Past Awards

- 2025 National South Korea Open Best Research Award
- 2025 RCAP Singapore Open Merit (Top 8) Award
- 2025 RCAP Abu Dhabi Influencer Award
- 2026 National South Korea Open Top 2 Award

Robot Design

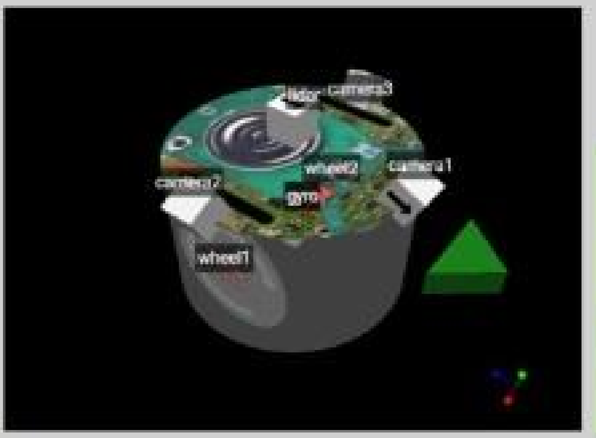
Motors
The robot uses two motors.
Allow the robot to move forward, turn, and adjust its path.

Cameras
The robot uses three cameras at the front, left, and right. They detect victims, hazardous signs, wall tokens, and floor types.

GPS
GPS measures the robot's coordinates and supports mapping, path planning, and return-to-start behavior.

Gyroscope
The gyroscope measures direction and rotation. It helps detect unexpected turning and supports stable movement.

LiDAR
LiDAR is placed at the top center of the robot and scans in 360 degrees. It detects obstacles, curved walls, and surrounding structures with accurate distance and angle data.

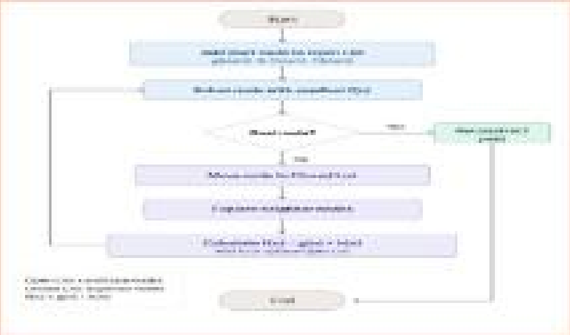


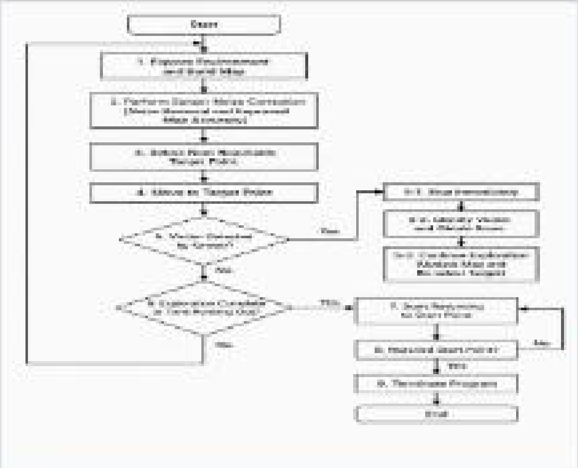
LiDAR provides obstacle and wall data. GPS and gyroscope provide position and direction data. Cameras provide victim, hazardous sign, wall token, and floor information. The mapper 2 combines these data into map layers, and the pathfinder uses the map to calculate routes. The executor controls the motors, while the state machine handles special events.

Software

Robot software combines navigation, detection, mapping, and control into one autonomous system. LiDAR, GPS, gyro, and camera data are processed in real time to update maps, detect objects, and decide movement.

Main tools: NumPy, OpenCV
Main algorithms: A*, BFS correction, HSV filtering, contour detection, ray casting





Mapping
The robot builds a real-time map using LiDAR, GPS, gyro, and camera data.
An expandable integer grid records walls, obstacles, floor types, explored areas, paths, and victims.
A granular grid improves environmental detail, especially for curved walls, and can be converted into the competition's Bonus Grid format.

Navigation
The robot explores the arena, chooses reachable targets, avoids obstacles, detects victims, and returns to the start when exploration ends or time is low.
For path planning, A* was selected instead of Dijkstra because it reaches targets more efficiently using:

$$f(n) = g(n) + h(n)$$

A navigation preference layer helps the robot avoid walls and risky areas.

Wall Token Detection
Front, left, and right cameras detect victims and hazard signs.
The system uses HSV color filtering, wall masking, contour analysis, bounding boxes, and ray casting to estimate token positions.
Victims and hazards are classified using pixel distribution patterns, allowing detection even with partial images.

Testing
Navigation, detection, and mapping were tested on multiple maps.
The robot was checked for target reaching, obstacle avoidance, blocked-path recovery, victim detection, and return-to-start behavior.
Testing showed that A*, color-based detection, and granular mapping were efficient and reliable in simulation.